Yangze Liu

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Education

University of Illinois Urbana-Champaign Aug 2021 – May 2025
B.S. in Liberal Arts and Sciences, Computer Science + Philosophy
GPA: 3.77/4.0

Research Interests

I study LLM-based agents, using controlled experiments to understand what actually drives their behavior and performance: where the knowledge in an agent's memory comes from, whether a model is coupled to its harness, whether a benchmark rewards the decisions it claims to test, and what goes wrong in an agent's actions even when it passes its task. I want to use this understanding to build better agents that can run safely in real environments. Earlier I worked on data selection for efficient training.

Publications

* Equal contribution.

Published and accepted

[1] **Yangze Liu** and Hong Liu. Rethinking Dataset Quantization: Efficient Coreset Selection via Semantically-Aware Data Augmentation. *Transactions on Machine Learning Research (TMLR)*, 2026. OpenReview.

[2] Yibing Liu*, **Yangze Liu***, Xiaolong Yin, Bin Wang, Chong Zhang, Hao Yin, and Zhongyi Han. OpenClawBench: Benchmarking Process-side Anomalies in Real-world Agent Execution Trajectories. *Advances in Neural Information Processing Systems (NeurIPS), Evaluations & Datasets Track*, 2026. Poster. arXiv:2605.29253.

Under review: LLM agents

[3] **Yangze Liu** and Zhongyi Han. What Does a Harness Buy? Tokens, Mostly. Under review at ICLR 2027. arXiv:2610.04433.

[4] **Yangze Liu** and Zhongyi Han. Experience Is Not Knowledge: Where the Decision Knowledge in an Agent's Memory Comes From. Under review at ICLR 2027.

[5] **Yangze Liu** and Zhongyi Han. Asking Earns Nothing: Scoring the Decision to Act in BFCL Multi-Turn. Under review at ICLR 2027. arXiv:2610.04429.

Research Experience

Shandong University Oct 2025 – Present
Full-time Research Assistant, advised by Prof. Zhongyi Han

- **Is a model coupled to its harness?** [3] Coding-agent harnesses ship new releases constantly and are credited with pass-rate gains, but whether the same model actually does better in a different harness had rarely been separated from run-to-run noise. Across five models run through Claude Code, mini-SWE-agent, and OpenCode on SWE-bench Verified, swapping the harness on a hard subset flipped as many tasks as rerunning the same one, and the only harness effect that cleared the noise was a loss. What the harness did decide was cost: up to three times per task for the same model.
- **Where the knowledge in an agent's memory comes from.** [4] Built a controlled Slay the Spire testbed that holds the player model, combat executor, prompt, and memory format fixed and varies only the source of the agent's memory, and designed the memory-construction methods it compares. The agent's self-written experience memory did not raise its clear rate, whereas memories holding correct, usable rules did, whether they came from simulator statistics, a stronger model's prior knowledge, or a stronger teacher reading the agent's own trajectories.
- **Scoring the decision to ask.** [5] Showed that the BFCL multi-turn scorer skips the turn on which an agent should ask for missing information, so asking earns no credit. Built a judge-free decision metric from the benchmark's own paired complete and incomplete requests. A one-line prompt pushing an agent toward action raised its official score by 13.5 to 23.5 points with no detectable change in its decisions, and the model that decided best ranked sixth of seven officially. Also checked the answer keys by hand and identified 31 incorrect ones.
- **OpenClawBench** (co-first author) [2]. A benchmark of process-side anomalies in real agent execution trajectories, where an agent passes the final task check but still ignores errors, makes unsafe external writes, or commits to weakly grounded claims. Built its dataset from generation through cleaning, and trained its trajectory-level anomaly detector, a LoRA-fine-tuned open model.

- **Auditing public agent leaderboards.** Audited HAL, τ-bench, and τ^2-bench as measurement instruments and reported four reproducible scoring and cost-accounting faults to their maintainers with reproduction scripts. In one, an agent on τ-bench Airline was graded in only 39% of its episodes, and grading every episode reordered the scaffolds for six of ten models. The HAL maintainers reproduced and confirmed three of the faults and added disclosures to the leaderboard.

Research collaboration with Prof. Hong Liu Jan 2024 – Jun 2026

Prof. Liu was then at Osaka University and is now at Xiamen University

Developed a coreset selection method that combines semantically aware data augmentation with a single randomly initialized network shared across augmentation, selection, and training. It matches or exceeds prior methods on ImageNet-1k and fine-grained datasets at lower cost; first-author paper in TMLR [1].

Other Manuscripts

All under review at ICLR 2027.

Data selection

[6] Are Coreset Selection Methods Worth Their Cost? **Y. Liu**, Z. Han. arXiv:2609.22894.

[7] When Hard-Example Selection Fails: Gaussian Thresholds and Empirical Diagnostics. **Y. Liu**, Z. Han.

[8] What Does 5% Mean? The Coreset Regime Boundary Is Pinned in Samples per Class. **Y. Liu**, X.-L. Yin, Z. Han.

[9] Useful to Whom? The Easy-versus-Coverage Boundary Moves with the Learner. **Y. Liu**, X.-L. Yin, Z. Han. arXiv:2610.00221.

Recursive training

[10] A Dominant Supplier Slows Recursive Drift More Than It Steers It. **Y. Liu**, Z. Han. arXiv:2609.11146.

[11] Break Step: Recursive Training Resonates with Replayed Sampling Noise. **Y. Liu**, Z. Han. arXiv:2609.11149.

Other Experience

University of Illinois Urbana-Champaign Oct 2023 – May 2025

Research Assistant, advised by Prof. Nishant Garg

Led development of Crystal Vision, a cross-platform VR and PC educational game built in Unreal Engine 5 and adopted in CEE 300 (Behavior of Materials) to teach crystallography. The project later received research funding for continued development.

Service

Reviewer, International Conference on Learning Representations (ICLR) 2027

Skills

Programming	Python, C/C++, Java
ML and LLM systems	PyTorch, vLLM (multi-node inference serving), LoRA fine-tuning
Agents and benchmarks	Claude Code, mini-SWE-agent, OpenCode, Harbor; SWE-bench, BFCL, τ-bench, GAIA
Infrastructure	Linux, multi-GPU experiment management across shared servers
Languages	Mandarin Chinese, English

Last updated October 13, 2026